ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/261251353

Paleomagnetic evidence for a Late Cretaceous deformation of the Great Valley Sequence, Sacramento Valley, California

Article · January 1978

CITATIONS
36

READS
61

1 author:



Edward A. Mankinen
United States Geological Survey
112 PUBLICATIONS **3,343** CITATIONS

SEE PROFILE

All content following this page was uploaded by Edward A. Mankinen on 17 April 2014.

The user has requested enhancement of the downloaded file.

Jour. Research U.S. Geol. Survey
Vol. 6, No. 3, May–June 1978, p. 383–390

PALEOMAGNETIC EVIDENCE FOR A LATE CRETACEOUS DEFORMATION OF THE GREAT VALLEY SEQUENCE, SACRAMENTO VALLEY, CALIFORNIA

by EDWARD A. MANKINEN, Menlo Park, Calif.

Abstract.—Paleomagnetic samples from five localities within the Great Valley sequence range in age from Late Jurassic to Late Cretaceous. All samples possess normal polarity, and alternating-field demagnetization experiments show that the remanence was acquired after the sequence was folded. A mean paleomagnetic pole position determined from 17 demagnetized samples is located at 72° N., 181° E., with the radius of the 95-percent circle of confidence equal to 5.4°. This pole position agrees very well with known Cretaceous poles from North America, a fact indicating that the deformation and remagnetization of the sequence must have occurred during the Late Cretaceous.

The Great Valley sequence of Bailey, Irwin, and Jones (1964) forms a thick north-trending belt of marine sedimentary rocks along the west margin of the Sacramento Valley. In general, the strata are homoclinical, with dips ranging from about 20° E. along Dry Creek near the north end of the sequence to as much as 70° E. along Putah Creek near the south end. The main rock types are mudstone, siltstone, sandstone, and conglomerate containing locally abundant fossils and representing a nearly continuous record of Late Jurassic to Late Cretaceous sedimentation. Detailed descriptions of the geology may be found in Bailey, Irwin, and Jones (1964), Ojakangas (1968), and Murphy, Rodda, and Morton (1969).

At the north end of the sequence, the strata unconformably overlie igneous and metamorphic rocks of the Klamath Mountains province (Irwin, 1960). Toward the west, they overlie mafic and ultramafic rocks of the Coast Range ophiolite (Bailey and others, 1970; Blake and Jones, 1977), and toward the east, the strata dip beneath upper Tertiary and Quaternary rocks.

Paleomagnetic samples were collected from five localities within the sequence (fig. 1). Samples from the Dry Creek–Budden Canyon section range in age from Hauterivian to early Coniacian (Murphy and others, 1969). Samples from the South Fork of Cottonwood Creek are of late Turonian through Coniacian age (Jones and Bailey, 1973). Samples from McCarty Creek are of Berriasian to early Hauterivian, and samples from Grindstone Creek are of mid-Tithonian to early Berriasian age (Jones and others, 1969; Imlay and Jones, 1970). The Venado Formation of Kirby (1943) sampled at Putah Creek is of Turonian age (Brown and Rich, 1960; Popenoe and others, 1960).

Acknowledgments.—Details of the geology and paleontology of the sequence, provided by D. L. Jones, were invaluable in the selection of sites for paleomagnetic sampling. C. S. Grommé contributed the samples from Putah Creek and many helpful discussions during the course of this study. Special thanks are also due A. H. Atkinson for assistance in the field and D. P. Elston for preliminary magnetic measurements of the collection.

METHODS USED

Samples were collected using a portable core drill and oriented using a magnetic compass. A total of 297 cores was collected, spanning a total stratigraphic thickness of approximately 15 000 m. Each core was cut into specimens 2.45 cm in diameter and 2.50 cm long, and one such specimen from each core was measured on a cryogenic magnetometer. Progressive AF (alternating-field) demagnetization experiments were performed using either a three-axis tumbler (Doell and Cox, 1976a) or a commercial 400-Hz (hertz) nontumbling demagnetizer. When the three-axis tumbler was used, the specimens were demagnetized twice at each increment of alternating field, and a vectorial method described by Grommé and Mankinen (1976) for eliminating the effects of rotational remanent magnetization was used to determine the true remanent magnetization at each step.

Weak-field susceptibilities were measured at 1000 Hz by using a commercial bridge. Saturation magnetizations were determined from magnetization curves obtained in fields from 0 to 9000 oersteds.

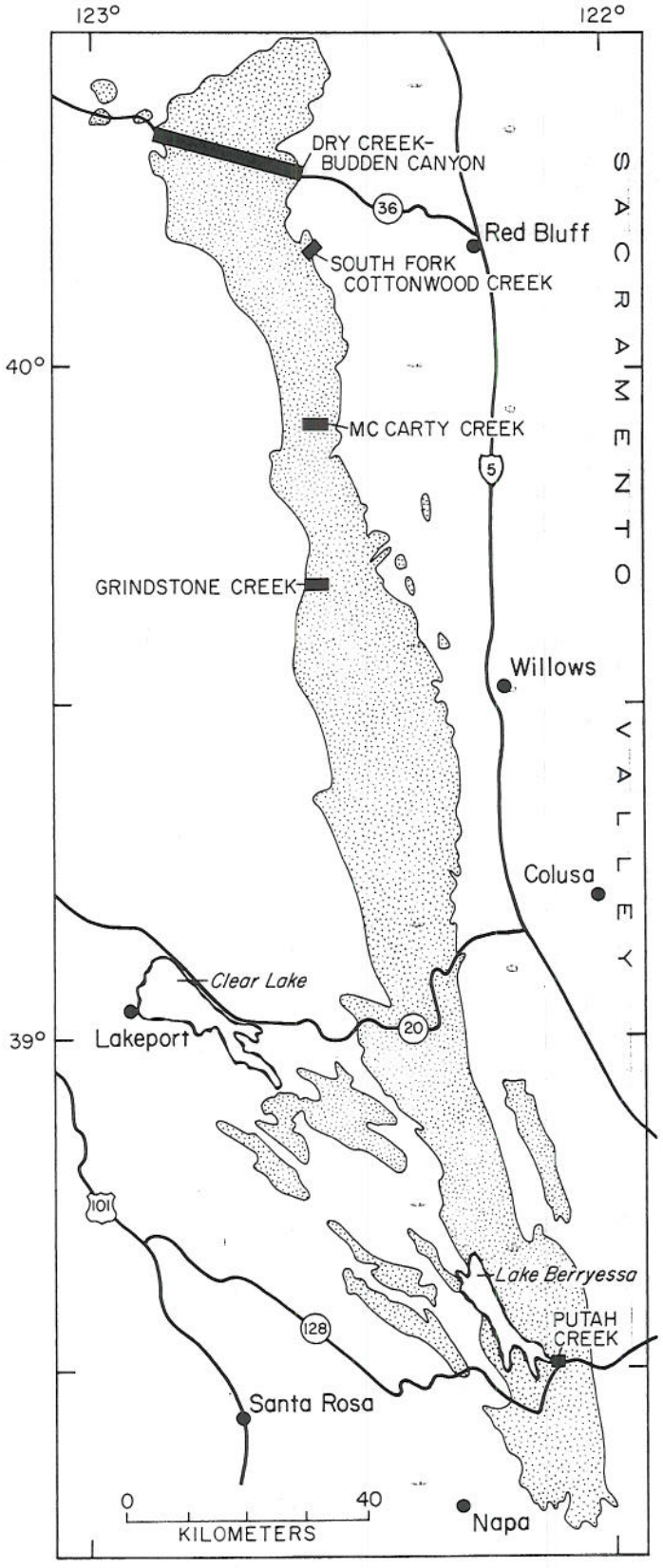


FIGURE 1.—Locations and informal names of sampled stratigraphic sections discussed in text. General distribution of Great Valley sequence (stippled) modified from Bailey, Irwin, and Jones (1964).

Saturation remanent magnetizations were induced in a field of 8000 Oe. Strong-field thermomagnetic curves were obtained using an automatic recording balance described by Doell and Cox (1967b). These experiments were made in air by using applied fields of 2500 to 4000 Oe at heating and cooling rates of 10°C/min.

REMANENCE MEASUREMENTS

NRM (natural remanent magnetization) measurements show only normal polarity for all samples. Average NRM directions and other magnetic properties were grouped by locality and are given in table 1. A mean direction of magnetization, calculated from the average for each locality, has a 95-percent-confidence circle (α_{95}) of 5° radius with an angular standard deviation to the mean of 5°. However, when all directions are corrected for the attitude of bedding, α_{95} increases to 16° and the angular standard deviation to 19°, demonstrating that the remanence was acquired after the sequence was folded.

In order to determine if the directions could be the result of VRM (viscous remanent magnetization) acquired in the present Earth's field, all samples except for those from Putah Creek were stored in the laboratory for 1 year before being remeasured. The average angular difference between measurements on any one sample was only 5°, a fact suggesting the absence of unstable components of NRM.

Progressive AF demagnetization experiments up to fields of 400 to 600 Oe were made on 19 samples from the four northern localities. These samples were selected from various stratigraphic levels throughout the sequence in order to get a representative sampling over a long period of time. When the logarithm of magnetization is plotted against the demagnetizing field, all the data fall on nearly straight lines, which again is an indication that significant components of anomalous magnetization are absent. The median demagnetizing field, the field at which the NRM intensity is reduced to half its initial value, ranged from 60 to 190 Oe with values most commonly between 100 and 150 Oe. All samples showed a systematic progression of direction along arcs of great circles, and the "best" direction for each was taken to be the last point that followed this systematic trend. The optimum demagnetizing field for all was between 150 and 350 Oe.

After demagnetization, all samples except two defined a group with a mean inclination of +65° and a declination of 339° (fig. 2). For these 17 samples, α_{95} was 4° and the precision parameter K of Fisher (1953) was 96. After the bedding correction was ap-

TABLE 1.—*Average NRM (natural remanent magnetization) directions and magnetic properties*

[N, number of samples; I, mean inclination of NRM, in degrees, positive downward; D, mean declination of NRM, in degrees east of north; u, uncorrected; c, corrected for attitude of bedding; J, geometric mean NRM intensity ($\times 10^{-5}$ emu/cm³); X, geometric mean susceptibility ($\times 10^{-4}$); Q, geometric mean in-place ratio of remanent to induced magnetization (Koenigsberger ratio); SD_J, SD_X, SD_Q, range of one standard deviation of geometric means for J, X, and Q, respectively; K, precision parameter of Fisher (1953); α_{95}, 95-percent confidence circle about average direction; and s.d., angular standard deviation about average direction]

Locality	N	Average NRM directions				J	SD_J	X	SD_X	Q	SD_Q
		I_u	D_u	I_c	D_c						
Budden Canyon	28	+71	007	+59	052	0.46	{1.35 .16	0.32	{1.01 .10	0.27	{0.59 .13
Dry Creek	95	+68	017	+58	050	1.20	{5.24 .28	1.31	{5.30 .32	.17	{.42 .07
South Fork Cottonwood Creek	10	+68	360	+35	034	1.97	{5.34 .73	1.27	{3.45 .46	.29	{.36 .24
McCarty Creek	41	+69	345	+38	069	.30	{.76 .12	.24	{.48 .12	.24	{.38 .15
Grindstone Creek	37	+68	337	+32	064	.43	{.82 .22	.23	{.32 .16	.36	{.64 .20
Putah Creek	73	+70	353	+20	073	2.18	{5.08 .94	5.23	{14.9 1.84	.08	{.14 .05
Average direction all localities	6	+69	356	+41	058						
Direction statistics		Uncorrected: K = 218, α_{95} = 5, s.d. = 5		Corrected: K = 18, α_{95} = 16, s.d. = 19							

plied, the mean inclination was +52° and the declination 46°, with $K = 18$ and $\alpha_{95} = 9°$. Clearly, even the stable magnetization recovered by AF demagnetization was acquired after the sequence was folded. Of the two samples that differed significantly from this group, one showed little change in direction during demagnetization, a fact indicating either that it became remagnetized at a later time than the other samples or that, because it was collected near a fault, some tectonic correction should have been applied. The second sample showed a progressive change toward a probable reversed direction, but no stable endpoint was reached before its magnetization became random. For this reason, it was not possible to determine whether the sample became remagnetized during a time when the Earth's field was reversed or whether some primary reversed magnetization was not completely destroyed during the postdeformation remagnetization.

Demagnetization curves for eight samples from Putah Creek also plot as straight lines, although the slopes are considerably steeper than those for the northern sections, a fact indicating that the NRM is softer at this locality. Median demagnetizing fields between 50 and 70 Oe and magnetic instability of most samples in fields as low as 100 to 150 Oe also suggest soft NRM. Changes in direction during demagnetization were much less systematic than in the other sections, and, as a result, the determination of the best direction for each sample tended to be more subjective. The mean direction of magnetization for these demagnetized samples, uncorrected for bedding attitude, has an inclination of +56° and a declination of 344° ($N = 8$, $K = 23$, $\alpha_{95} = 12°$), which is not statistically different from the mean calculated for the northern sections. Applying the tectonic correction gives an inclination of +26° and a declination of 58°, which corresponds to no known ancient field direction, again demonstrating remagnetization after deformation. The greater scatter in directions at this locality may simply be the result of the less stable magnetization, although more than one episode of remagnetization is certainly a possibility.

MAGNETIC PROPERTIES

The most obvious differences in magnetic properties between sections is that the Putah Creek samples have much higher susceptibilities and lower Koenigsberger ratios (table 1). Of the 73 samples from this locality, only 6 have susceptibilities in the 10^{-5} range with the remainder being 2×10^{-4} and higher. Of the samples from the northern sections, only 15 percent have susceptibilities approaching values as high as those from Putah Creek. Strong-field thermomagnetic curves were obtained from six samples taken from various

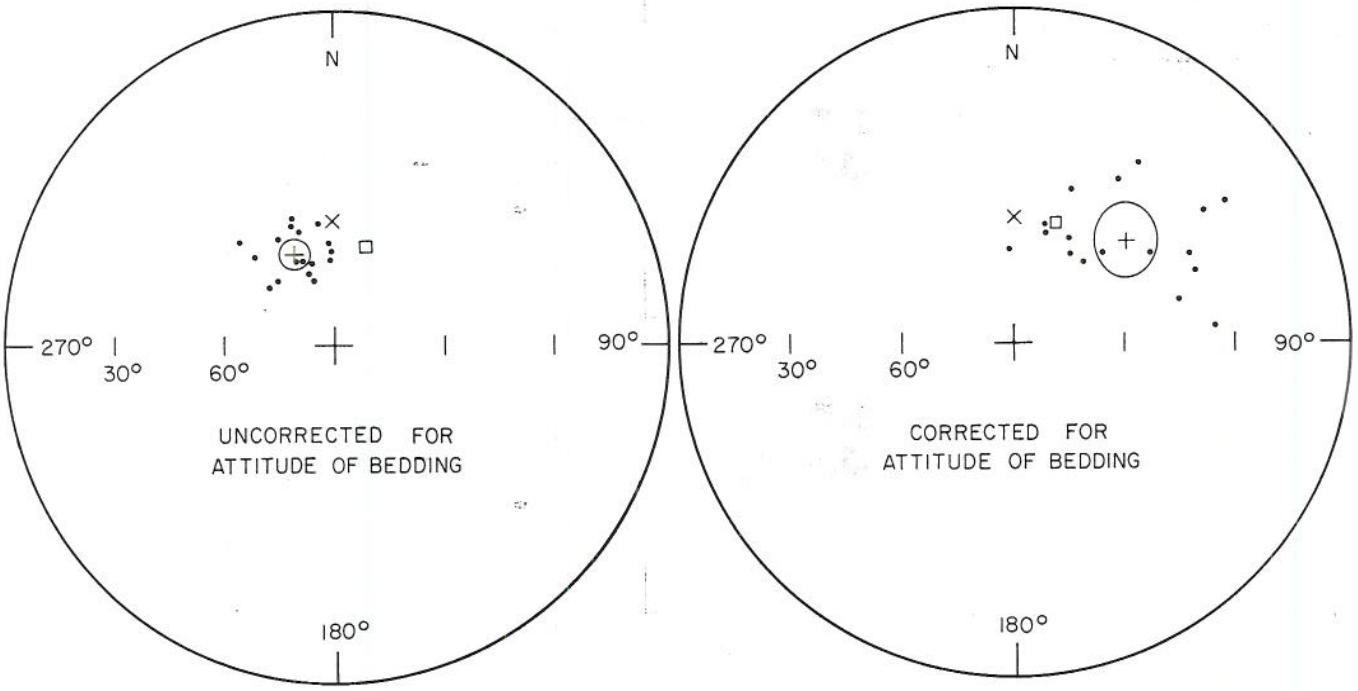


FIGURE 2.—Average remanent magnetization directions (dots) after magnetic cleaning. All directions on lower hemisphere of equal-area projection. Small cross, mean of all directions with its associated 95-percent-confidence circle; ×, ideal dipole field direction; small square, present field direction.

levels throughout the sequence and all have Curie temperatures between 575° and 580°C, facts showing that the main magnetic mineral present is magnetite.

Given identical mineralogy, both the susceptibility and the M_S (saturation magnetization) of a sample will vary with the volume fraction of magnetic minerals present, and the susceptibility will also vary with changes in grain size. Since both quantities increase with volume fraction, there should be a linear relation between saturation magnetization and susceptibility if this is the only variable. As shown in table 2 and figure 3*A*, the increase in both properties is not entirely linear, a fact suggesting some variation in grain size as well.

TABLE 2.—*Variation of magnetic properties with susceptibility*

[X, susceptibility ($\times 10^{-4}$); M_{RS}, saturation remanence ($\times 10^{-3}$ emu/g); M_S, saturation magnetization (emu/g); N.d., not determined]

	Putah Creek			Northern localities		
X	M_{RS}	M_S	M_{RS}/M_S	M_{RS}	M_S	M_{RS}/M_S
12.2	7.35	0.706	0.010			
7.95	3.60	.460	.008			
6.21	3.07	.368	.008			
5.65	4.60	.339	.014			
5.53				8.93	0.209	0.043
4.94	3.33	.237	.014			
4.34				5.18	.107	.048
3.47				8.53	.241	.035
.97				2.57	.051	.050
.53				1.50	.013	.115
.44	.88	.012	.073			
.33	.40	N.d.	N.d.			
.33				1.16	.009	.129
.33				1.13	.006	.188
.32				.81	N.d.	N.d.
.11				.25	N.d.	N.d.
.11				.59	.002	.295

The saturation remanence (M_{RS}) of a sample varies with volume fraction and also with the coercive force of the minerals, which is dependent on grain size. If the susceptibility differences seen are simply due to changes in concentration, then the saturation remanence at any given susceptibility should be the same if the coercive force is constant. Where there is an overlap in susceptibilities between sections (table 2), the saturation remanence of samples from Putah Creek is considerably lower, indicating a decrease in coercive force that must be due to an increase in grain size.

Since, for a constant volume fraction of magnetite, increase in grain size reduces the saturation remanence without changing the saturation magnetization, perhaps a more reliable indication of grain size variation would be the ratio of the two properties. This ratio in all samples is very small (table 2), indicating that the remanence is carried by multidomain grains; Stacey

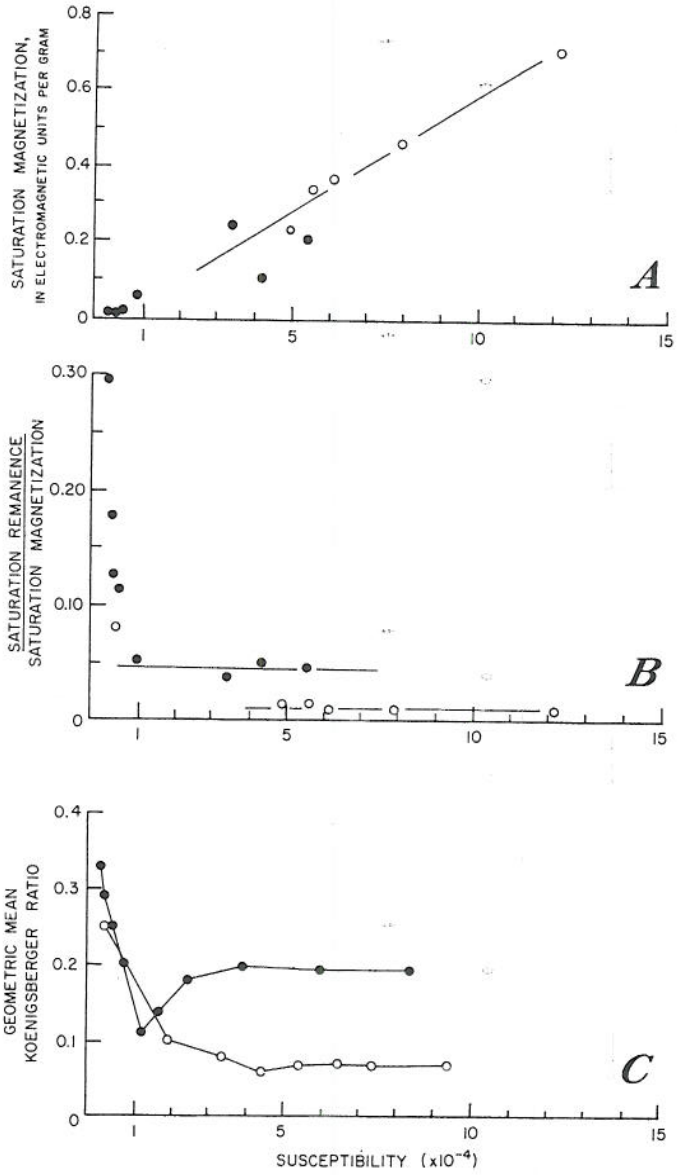


FIGURE 3.—Variation of magnetic properties with susceptibility. *A*, Saturation magnetization. *B*, Ratio of saturation remanence to saturation magnetization. *C*, Geometric mean Koenigsberger ratio. Circles, samples from Putah Creek; dots, samples from northern localities.

and Banerjee (1974) have shown that the saturation remanence should be about one-half of the saturation magnetization for single-domain grains. A plot of the ratio M_{RS}/M_S against susceptibility (fig. 3*B*) indicates a large variation in grain size throughout the sequence. With the exception of a few low-susceptibility samples that exhibited demagnetization behavior similar to those from the northern localities, the ratio for the Putah Creek samples is much lower, suggesting an appreciably larger grain size. This interpretation is substantiated in a general way by the lower Koenigsberger ratios for the Putah Creek samples (fig. 3*C*), because this ratio tends to increase with a decrease in grain size. These experiments all suggest that the remanence of the Putah Creek samples is carried by larger multidomain grains than are found in samples from the sections to the north. A larger grain size would be the most probable cause of the relatively unstable magnetization in the samples from Putah Creek.

DISCUSSION

A VGP (virtual geomagnetic pole) was calculated for each sample demagnetized (uncorrected for bedding attitude), except for those from Putah Creek, and the VGP's were then averaged to obtain a mean paleomagnetic pole located at 72° N., 181° E. ($N=17$, $K=44$, $\alpha_{95}=5.4°$). This pole is not significantly different from known reliable Cretaceous poles from North America (table 3 and fig. 4), a fact suggesting that the Great Valley sequence must have been folded and remagnetized during the Late Cretaceous. Pole positions omitted from the list of reliable Cretaceous poles in table 3 are those for the Dakota Sandstone (Runcorn, 1956) because of insufficient sampling and for the Bucks batholith and Guadaloupe Mountains (Grommé and others, 1967) because of their probable Late Jurassic age. Poles determined for the Boulder

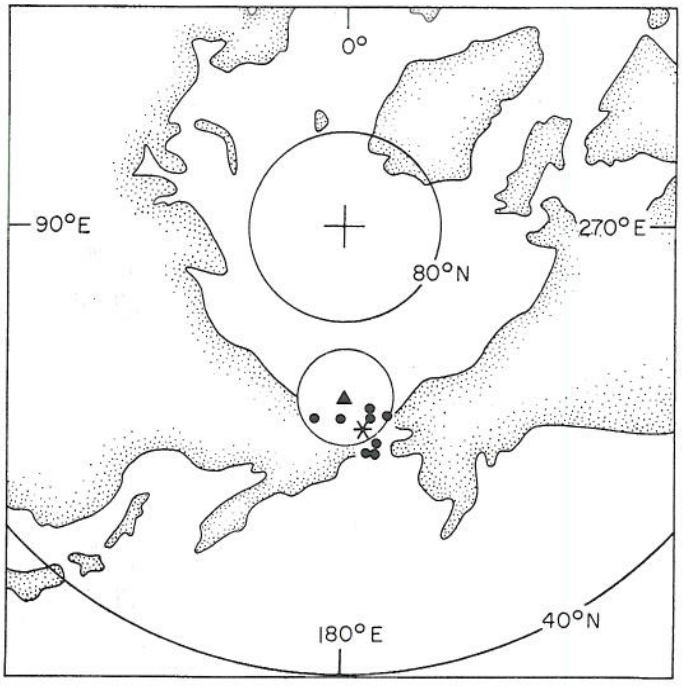


FIGURE 4.—Equal-area projection of part of northern hemisphere showing the pole position determined for the Great Valley sequence (triangle) and its associated 95-percent-confidence circle. Dots are Cretaceous pole positions listed in table 3; average of these poles shown as an asterisk.

TABLE 3.—*Cretaceous paleomagnetic pole positions for North America*

Rock unit	Locality	Pole position		Reference
Mt. Megantic intrusive	Quebec	69°N	172°E	Larochelle (1961)
Isachsen diabase	Canadian Arctic	69°N	180°E	Larochelle and Black (1963) Larochelle, Black and Wanlass (1965)
Sierra Nevada plutons	California	69°N	195°E	Grommé and Merrill (1965)
Mt. Ascutney intrusive	Vermont	65°N	186°E	Opdyke and Wensink (1966)
Elkhorn Mountains Volcanics	Montana	69°N	189°E	Hanna (1967)
Monteregian Hills intrusives	Quebec	70°N	189°E	Larochelle (1969)
Alkalic intrusives	Arkansas	65°N	187°E	Scharon and Hsu (1969)
Niobrara Formation	Colorado, Kansas	66°N	189°E	Shive and Frerichs (1974)
	Average:	68°N	186°E	

batholith (Hanna, 1973) and the Mesaverde Group (Kilbourne, 1969) were omitted because of their large error limits, and the pole for the Howe Sound plutons (Symons, 1973), because of its anomalous direction. Symons' high pole latitude (83° N.) for the Howe Sound plutons suggests that the plutons may have acquired their magnetization considerably later than the time of emplacement or, alternately, that some tectonic rotation has been involved, because they are located in the western cordillera, where anomalous directions have sometimes been observed (Beck, 1975). The pole for the Niobrara Formation was calculated from Shive and Frerichs' (1974) published data without the Kemmerer and Centennial Valley sites because of the large error limits for data from those localities.

NRM directions from Grindstone Creek and the lower 2500 m of the McCarty Creek section are tightly grouped, suggesting that the original remagnetization has not been much modified; therefore, a VGP was calculated for each section for comparison. The VGP for Grindstone Creek is located at 71° N., 189° E. ($N = 37$, $K = 29$, $\alpha_{95} = 4.5°$), and for McCarty Creek at 74° N., 196° E. ($N = 29$, $K = 56$, $\alpha_{95} = 3.6°$), neither of which is significantly different from the mean calculated for the demagnetized samples nor from the known Cretaceous poles.

Whereas the pole position determined from the Great Valley sequence is completely consistent with other poles determined for the Cretaceous Period, the possibility that the pole may have remained at the same location until some time during the Tertiary must be considered. Studies summarized by McElhinny (1973) have shown that the pole for the late Tertiary of North America is located near the geographic pole, but there are no reliable data for the early Tertiary. One method of determining an early Tertiary pole position would be to locate a well determined pole of the same age from Europe or Africa. The finite-rotation method of Pitman and Talwani (1972) could then be used to obtain the relative paleogeographic positions of the continents and the pole position relative to North America. A paleomagnetic pole ideal for this purpose is that given by Irving (1964) for the igneous complexes of the British Isles; recent age determinations (Duncan and others, 1972) show this pole to be Paleocene. By holding North America fixed and rotating Europe toward North America about Pitman and Talwani's (1972) 63-million-year

pole of rotation, a Paleocene paleomagnetic pole relative to North America would be located at 78° N., 170° E. This Paleocene pole is 11° away from the average Cretaceous pole (fig. 5) and on a very nearly direct path toward the Oligocene pole at 78° N., 146° E., reported by Grommé and McKee (1971), a fact strongly suggesting that the pole determined for the Great Valley sequence is indeed Cretaceous in age. This interpretation is in excellent agreement with suggestions by Irwin (1964), Blake, Irwin, and Coleman (1967), and Ernst (1970) that thrust faulting along the Great Valley province boundary occurred during the Late Cretaceous. However, this faulting may represent only the beginning of deformation in this region, because data presented by Hackel (1966) suggest that uplift of the sequence at the south end of the Sacramento Valley did not occur until after the early Eocene. Unfortunately, the error limits for the Putah Creek samples are too large for the data to be useful in proving or disproving this suggestion.

Considering the great sedimentary thicknesses involved, it is surprising that all samples seem to be of normal polarity. This suggests either that the remagnetization took place over a relatively short period of time or that the long Cretaceous normal polarity interval may have persisted until quite late in the Cretaceous. It is not now clear which alternative is the most likely. Studies of oceanic magnetic anomalies (Larson and Pitman, 1972) suggest that the Cretaceous normal polarity interval ended during the Coniacian Stage, and volcanic rocks having apparently the same age but mixed polarities (Hanna, 1967) tend to support this idea. However, other studies such as those by Pechersky and Khramov (1973), Shive and Frerichs (1974), and Keating, Helsley, and Pessagno (1975) show predominantly normal polarities at least until some time during the Maestrichtian.

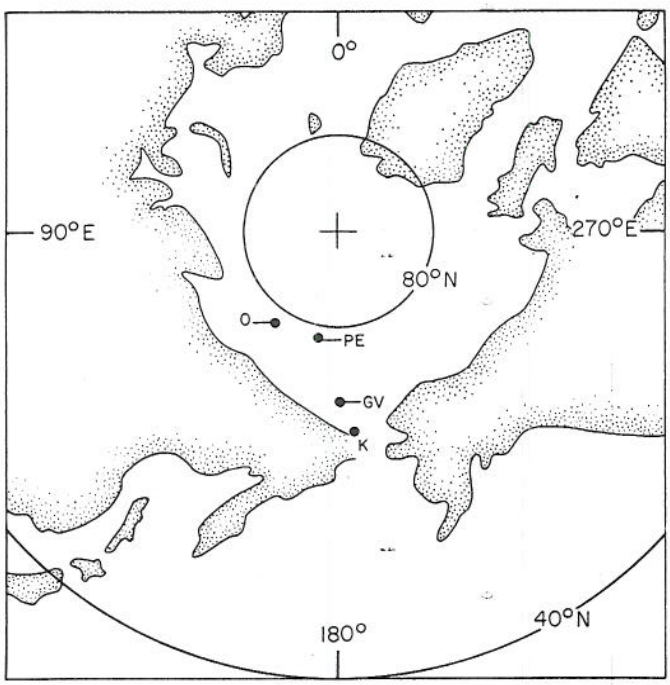


FIGURE 5.—Equal-area projection of part of the northern hemisphere showing the average Cretaceous pole (K) from table 3 and the poles for the Great Valley sequence (GV), the rotated European Paleocene (PE), and the Oligocene (O), from Grommé and McKee (1971).

The question of why these strata became remagnetized still remains to be answered. The deformation and remagnetization of the sequence are closely related in time, but it seems unlikely that the deformation itself could be the cause. The strata are only gently tilted throughout much of the sequence, and in none of the areas sampled are they penetratively deformed. Perhaps the remagnetization is due to the acquisition of a viscous PTRM (partial thermoremanence) similar to that described by Pullaiah, Irving, Buchan, and Dunlop (1975). If the strata were subjected to slightly elevated temperatures (100°C or more) for a long period of time, it is possible that some of the original NRM could be destroyed and a viscous PTRM produced, which would then become stabilized during cooling. To test this possibility, thermal demagnetization experiments were performed on 10 samples from the sequence. Changes in direction for two samples indicated that their primary magnetization may have been that of reversed polarity, but results from the other samples showed little or no improvement over those obtained by AF demagnetization. Both the thermal and AF demagnetization experiments suggest that the remagnetization of the sequence was very nearly complete and that neither technique is likely to provide results that are not ambiguous.

REFERENCES CITED

Bailey, E. H., Blake, M. C., Jr., and Jones, D. L., 1970, On-land Mesozoic oceanic crust in California Coast Ranges: U.S. Geol. Survey Prof. Paper 700–C, p. C70–C81.

Bailey, E. H., Irwin, W. P., and Jones, D. L., 1964. Franciscan and related rocks and their significance in the geology of western California: California Div. Mines and Geology Bull. 183, 177 p.

Beck, M. E., Jr., 1975, Remanent magnetism of the Twin Sisters dunite intrusion and implications for the tectonics of the western cordillera: Earth and Planetary Sci. Letters, v. 26, p. 263–268.

Blake, M. C., Jr., Irwin, W. P., and Coleman, R. G., 1967, Upside-down metamorphic zonation, blueschist facies, along a regional thrust in California and Oregon: U.S. Geol. Survey Prof. Paper 575–C, p. C1–C9.

Blake, M. C., Jr., and Jones, D. L., 1977, Plate tectonic history of the Yolla Bolly junction: Geol. Soc. America Guidebook, Cordilleran Sec. Mtg., Sacramento, Calif., 14 p.

Brown, R. D., Jr., and Rich, E. I., 1960, Early Cretaceous fossils in submarine slump deposits of late Cretaceous age, northern Sacramento Valley, California: U.S. Geol. Survey Prof. Paper 400–B, p. B318–B320.

Doell, R. R., and Cox, Allan, 1967a, Analysis of alternating field demagnetization equipment, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 241–253.

——— 1967b, Recording magnetic balance, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 440–444.

Duncan, R. A., Petersen, Nikolai, and Hargraves, R. B., 1972, Mantle plumes, movement of the European plate and polar wandering: Nature v. 239, p. 82–86.

Ernst, W. G., 1970, Tectonic contact between the Franciscan mélange and the Great Valley sequence—Crustal expression of a late Mesozoic Benioff zone: Jour. Geophys. Research, v. 70, p. 886–901.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. [London] Proc., Ser. A, v. 217, p. 295–305.

Grommé, C. S., and McKee, E. H., 1971, Mid-Tertiary paleomagnetic pole from volcanic rocks in the western United States: Am. Geophys. Union Trans., v. 52, p. 187.

Grommé, C. S., and Mankinen, E. A., 1976, Natural remanent magnetization, magnetic properties, and oxidation of titanomagnetite in basaltic rocks from DSDP Leg 34, *in* Yeats, R. S., Hart, S. R., and others, eds., Initial reports of the Deep Sea Drilling Project, v. 34: Washington, D.C., U.S. Govt. Printing Office, p. 485–494.

Grommé, C. S., and Merrill, R. T., 1965, Paleomagnetism of late Cretaceous plutons in the Sierra Nevada, California—Further results: Jour. Geophys. Research, v. 70, p. 3407–3420.

Grommé, C. S., Merrill, R. T., and Verhoogen, John, 1967, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California, and its significance for polar wandering and continental drift: Jour. Geophys. Research, v. 72, p. 5661–5684.

Hackel, Otto, 1966, Summary of the geology of the Great Valley, *in* Bailey, E. H., ed., Geology of Northern California: California Div. Mines and Geology Bull. 190, p. 217–238.

Hanna, W. F., 1967, Paleomagnetism of upper Cretaceous volcanic rocks of southwestern Montana: Jour. Geophys. Research, v. 72, p. 595–610.

——— 1973, Paleomagnetism of the late Cretaceous Boulder batholith, Montana: Am. Jour. Sci., v. 273, p. 778–802.

Imlay, R. W., and Jones, D. L., 1970, Ammonites from the *Buchia* zones in northwestern California and southwestern Oregon: U.S. Geol. Survey Prof. Paper 647–B, p. B1–B59.

Irving, E., 1964, Paleomagnetism and its application to geological and geophysical problems: New York, John Wiley and Sons, 399 p.

Irwin, W. P., 1960, Geologic reconnaissance of the northern Coast Ranges and Klamath Mountains, California, with a summary of the mineral resources: California Div. Mines Bull. 179, 80 p.

——— 1964, Late Mesozoic orogenies in the ultramafic belts of northwestern California and southwestern Oregon: U.S. Geol. Survey Prof. Paper 501–C, p. C1–C9.

Jones, D. L., and Bailey, E. H., 1973, Preliminary biostratigraphic map, Colyear Springs quadrangle, California: U.S. Geol. Survey Misc. Field Studies Map MF–517, scale 1:48,000.

Jones, D. L., Bailey, E. H., and Imlay, R. W., 1969, Structural and stratigraphic significance of the *Buchia* zones in the Colyear Springs-Paskenta area, California: U.S. Geol. Survey Prof. Paper 647–A, p. A1–A24.

Keating, B. H., Helsley, C. E., and Pessagno, E. A., Jr., 1975, Late Cretaceous reversal sequence: Geology, v. 3, p. 73–76.

Kilbourne, D. E., 1969, Paleomagnetism of some rocks from the Mesaverde Group, southwestern Wyoming and northeastern Utah: Geol. Soc. America Bull., v. 80, p. 2069–2074.

Kirby, J. M., 1943, Upper Cretaceous stratigraphy of the west side of Sacramento Valley south of Willows, Glen County, California: Am. Assoc. Petroleum Geologists Bull., v. 27, p. 279–305.

Larochelle, Andre, 1961, Application of paleomagnetism to geological correlations: Nature, v. 192, p. 37–39.

——— 1969, Paleomagnetism of the Monteregian Hills—Further new results: Jour. Geophys. Research, v. 74, p. 2570–2575.

Larochelle, Andre, and Black, R. F., 1963, An application of paleomagnetism in estimating the age of rocks: Nature, v. 198, p. 1260–1262.

Larochelle, Andre, Black, R. F., and Wanless, R. K., 1965, Paleomagnetism of the Isachsen diabasic rocks: Nature, v. 208, p. 179.

Larson, R. L., and Pitman, W. C., III, 1972, World-wide correlation of Mesozoic magnetic anomalies, and its implications: Geol. Soc. America Bull., v. 83, p. 3645–3662.

McElhinny, M. W., 1973, Paleomagnetism and plate tectonics: Cambridge, Cambridge Univ. Press, 358 p.

Murphy, M. A., Rodda, P. V., and Morton, D. M., 1969, Geology of the Ono quadrangle, Shasta and Tehama Counties, California: California Div. Mines and Geology Bull. 192, 28 p.

Ojakangas, R. W., 1968, Cretaceous sedimentation, Sacramento Valley, California: Geol. Soc. America Bull., v. 79, p. 973–1008.

Opdyke, N. D., and Wensink, Hans, 1966, Paleomagnetism of rocks from the White Mountain plutonic-volcanic series in New Hampshire and Vermont: Jour. Geophys. Research, v. 71, p. 3045–3051.

Pechersky, D. M., and Khramov, A. N., 1973, Mesozoic paleomagnetic scale of the USSR: Nature, v. 244, p. 499–501.

Pitman, W. C., III, and Talwani, Manik, 1972, Sea-floor spreading in the North Atlantic: Geol. Soc. America Bull., v. 83, p. 619–646.

Popenoe, W. P., Imlay, R. W., and Murphy, M. A., 1960, Correlation of the Cretaceous formations of the Pacific Coast (United States and northwestern Mexico): Geol. Soc. America Bull., v. 71, u. 1491–1540.

Pullaiah, G., Irving, E., Buchan, K. L., and Dunlop, D. J., 1975, Magnetization changes caused by burial and uplift: Earth and Planetary Sci. Letters, v. 28, p. 133–143.

Runcorn, S. K., 1956, Paleomagnetic survey in Arizona and Utah—Preliminary results: Geol. Soc. America Bull., v. 67, p. 301–316.

Scharon, LeRoy, and Hsu, I-Chi, 1969, Paleomagnetic investigation of some Arkansas alkalic igneous rocks: Jour. Geophys. Research, v. 74, p. 2774–2779.

Shive, P. N., and Frerichs, W. E., 1974, Paleomagnetism of the Niobrara Formation in Wyoming, Colorado, and Kansas: Jour. Geophys. Research, v. 79, p. 3001–3007.

Stacey, F. D., and Banerjee, S. K., 1974, The physical principles of rock magnetism: Amsterdam, Elsevier, 195 p.

Symons, D. T. A., 1973, Concordant Cretaceous paleolatitudes from felsic plutons in the Canadian Cordillera: Nature Phys. Sci., v. 241, p. 59–61.

View publication stats